Provenance-Aware, Patient-Disjoint Evaluation of Ophthalmic Screening with Controlled Document Reliability

Academic prototype — not a medical device, not clinically validated

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***Abstract*—Four-class fundus-photograph benchmarks for cataract, diabetic retinopathy, glaucoma, and normal screening are widely reported at 95–96% accuracy. This paper asks how much of that number reflects retinal pathology and how much reflects two properties of how such benchmarks are usually built: the acquisition source an image came from, and whether the same patient's fellow eye appears in both the training and test sets. We audit a merged four-class benchmark built on ODIR-5K, quantify both effects, and re-evaluate a standard image classifier under a corrected, patient-disjoint, deduplicated, provenance-aware protocol. We find that acquisition provenance alone predicts class with 66.7% accuracy using no pixels, that patient-level leakage produces no measurable inflation on a frozen linear classifier but produces a significant $+0.022$ macro-F1 inflation (McNemar $p < 0.001$) on a fine-tuned ResNet50 — the model that would actually be deployed — and that this effect grows monotonically with a model's capacity to memorize. We further show the trained classifier does not transfer to three of four unseen acquisition sources, and that its accuracy survives explicit age and sex matching, indicating it is not exploiting the demographic confounds present in the data. Building on this audit, we construct a screening prototype in which a disease prediction is produced from the fundus image alone by construction, while a companion referral document is used only to inform a deferral decision. We show raw OCR confidence is a poor predictor of extraction correctness (AUROC 0.52–0.64) but a learned estimator recovers this signal almost completely (AUROC 0.98–0.99), that document evidence roughly halves the area under the risk-coverage curve for deferral, and that — under a falsification rule fixed before the experiment — a learned deferral gate did not outperform two hand-written rules, a negative result we report rather than conceal. Our contribution is empirical throughout: an audit methodology, its quantified findings on a specific benchmark, and a working prototype whose safety property (image-only diagnosis) is enforced structurally rather than by convention.**

***Index Terms*—dataset bias, patient leakage, selective prediction, ophthalmic screening, document reliability, evaluation protocol**

I. Introduction

Ophthalmic screening papers built on public four-class fundus benchmarks routinely report accuracies in the 95–96% range using architectures such as EfficientNetB3 and DenseNet169. These numbers are usually taken as evidence that distinguishing cataract, diabetic retinopathy (DR), glaucoma, and normal fundus images is close to solved by modern convolutional networks. What such numbers measure is examined less often. A benchmark assembled by merging several independently curated sources into one folder per class carries structure beyond the images themselves: which camera or export pipeline produced a file, what resolution it was saved at, and whether a given patient's two eyes were split across train and test. None of these properties is retinal pathology, and a sufficiently capable model can exploit any of them.

This paper does not assert that the field's numbers are wrong. It asks, directly and measurably: of the reported performance on a four-class benchmark, how much is acquisition provenance, how much is patient-level memorization, and how much survives once both are controlled for? A screening tool that must answer on every image, regardless of its actual certainty, is in addition not a system a clinician could reasonably deploy — that a system should be able to decline to answer is an argument this paper makes, not a result it measures, and the two are kept separate throughout.

The paper proceeds in three parts. Section III audits a merged four-class benchmark and quantifies acquisition-provenance confounding and patient-level leakage. Section IV re-evaluates a standard classifier under a patient-disjoint, deduplicated, provenance-aware protocol, confirming the leakage finding on a fine-tuned deployable model rather than a linear proxy. Sections V–VI present a working screening prototype in which a referral document improves the decision to defer a case without ever influencing the diagnosis, and show that the reliability of a machine-read document must itself be estimated rather than assumed. Every research question below carries a falsification criterion fixed before the corresponding experiment was run; where an experiment produced a negative result, it is reported as such.

II. Related Work

This work applies existing audit and selective-prediction methodology to a specific benchmark and a specific prototype; it does not propose a new method in any of the areas below.

TABLE I
RESEARCH QUESTIONS AND WHERE EACH IS ANSWERED

RQ	Question
RQ1	How much of reported performance is acquisition provenance?
RQ2	How much is patient-level leakage and duplication?
RQ3	Is raw OCR confidence a usable reliability signal for extracted clinical fields?
RQ4	How does referral-document quality propagate into triage decisions?
RQ5	Can document evidence improve deferral without changing the diagnosis?

Provenance and dataset bias. Torralba and Efros [1] showed that a classifier can identify which benchmark an image was drawn from — "name that dataset" — using low-level image statistics alone. Liu and He [2] revisit this experiment a decade later on modern large-scale benchmarks and show the effect persists even for diverse, web-scale datasets. The provenance probe in this paper (Section III) is a direct application of that design to a four-class ophthalmic benchmark; it does not propose a new detection method.

Leakage inflation. Tampu *et al.* [3] documented that improper, non-patient-disjoint splitting inflates reported accuracy by 5–30 percentage points across several OCT classification benchmarks. This paper follows the same template for fundus photography and adds a capacity-sweep design (Section IV-B) to show that the size of the inflation depends on whether the model being evaluated can actually memorize a leaked example.

Selective prediction in ophthalmology. Leibig *et al.* [4] and DR|GRADUATE [5] established that a diabetic-retinopathy classifier benefits from an explicit option to abstain rather than answer on every case. This paper adopts that principle but adds a second evidence source — a referral document — under the constraint that the document must not be able to change the diagnosis, which neither prior system needed to address since neither incorporated a second modality.

Reliability-weighted fusion. Quality-aware Multimodal Fusion (QMF) [6], Trusted Multi-view Classification (TMC) [7], and MultiRetNet [8] for retinal imaging specifically motivate the deferral architecture in Section VI. Our instantiation of that idea is narrower: reliability is estimated for a document-extraction pipeline rather than a learned multimodal encoder, and the resulting comparison against hand-written rules is reported honestly when the learned version does not win.

Attribution evaluation in ophthalmology. Yavari *et al.* [9] evaluate attribution methods in retinal imaging and establish the standard this paper follows for its own explainability check (Section VI): attribution concentration and a model-randomization sanity check, without claiming that attribution identifies a clinical lesion absent expert annotation.

Document information extraction. DocILE [10] established the convention of evaluating document-extraction systems on layouts held out from development, which this paper follows directly for its synthetic referral corpus (Section V): extractors are tuned on three layouts and evaluated on three sealed ones opened only once.

TABLE II
DATASET COMPOSITION

Class	Images	Patients	Multi-finding	Mean age
Cataract	293	203	10.6%	66.8
DR	1,608	1,027	13.9%	55.7
Glaucoma	284	182	23.2%	63.0
Normal	2,873	1,852	2.0%	57.2

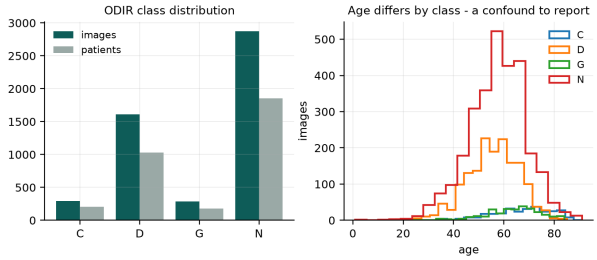


Fig. 1. Dataset composition by class and acquisition source.

III. DATA AND AUDIT

A. Dataset Composition

The evaluation dataset is the four-class subset of ODIR-5K: 5,058 images from 2,910 patients across cataract (C), diabetic retinopathy (D), glaucoma (G), and normal (N). Table II summarizes composition; Fig. 1 gives the full distribution.

Of 2,910 patients, 2,148 contribute both eyes, which is the structural precondition for patient-level leakage: if eyes rather than patients are the unit of splitting, a model can encounter one eye at training time and the fellow eye at test time. 354 patients carry different diagnoses in each eye, so splitting is performed by patient while stratification for class balance is performed at eye level. The corpus spans 91 distinct image resolutions with no EXIF metadata, meaning acquisition source cannot be read from a clean field and must instead be inferred (Section III-B).

Two confounds are reported here rather than deferred to a limitations section, because Section IV-E tests them directly: cataract patients average approximately 11 years older than diabetic-retinopathy patients, sex distribution differs by class, and 23.2% of glaucoma-labelled rows name a second condition in the original annotation. ODIR is less confounded than a naively merged benchmark, but it is not confound-free.

B. Provenance Audit

Acquisition source was assigned from file properties alone — extension, resolution, and pixel-level export artifacts — never from the class folder a file was drawn from, since labelling source from class would make any downstream probe circular. Three probes were run.

TABLE III
ANTI-CIRCULARITY CONTROL (E0C-II)

Source	n	Baseline	Within-source	Lift
`odir_jpg512`	1,678	0.640	0.644	+0.004
`crop_jpg256`	1,240	0.516	0.736	+0.219
`png_highres`	201	0.497	0.781	+0.284

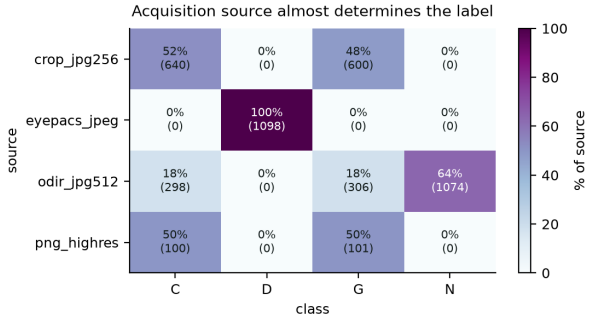


Fig. 2. Source × class heatmap.

E0a predicted class from file metadata only, with no pixel content: a majority-class baseline of 26.0% rose to **66.7%**. Every diabetic-retinopathy image in this benchmark is a `.jpeg` file and no other class contains a `.jpeg`, making file extension alone a perfect DR detector independent of image content.

E0b predicted acquisition source from FOV-normalized pixels (0.398 → 0.749), and E0c-i predicted source from image geometry alone (0.398 → 0.702): source leaves a detectable signature even after cropping to the retinal field of view and even when only crop shape is available.

E0c-ii, the anti-circularity control, held source constant and asked whether class could still be predicted from pixels within a single source. If provenance were doing all the work, this signal should vanish once source is fixed.

The result is asymmetric, and both halves are reported. Within `odir_jpg512` — the source containing the DR and normal classes — the signal essentially vanishes (+0.004), consistent with file extension having already explained nearly all separable variance for those two classes. Within `crop_jpg256` and `png_highres` — sources split roughly evenly between cataract and glaucoma — a real signal survives (+0.219 and +0.284), indicating the cataract-versus-glaucoma distinction is not purely a provenance artifact. **Provenance explains the diabetic retinopathy and normal classes almost entirely, and comparatively little of the cataract/glaucoma separation.** Fig. 2 and Fig. 3 carry this finding visually.

C. Deduplication

Two candidate detectors were evaluated against a ground truth of 1,654 known duplicate pairs. An ImageNet ResNet50 embedding, compared by cosine similarity, was tried first and **failed**: at an operating threshold of 0.999 it recovered **0%** of known duplicates, since fundus images are visually homogeneous to begin with and true duplicates differ by a

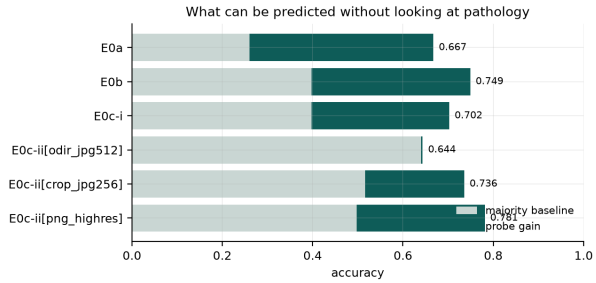


Fig. 3. Provenance probe comparison (E0a/E0b/E0c).

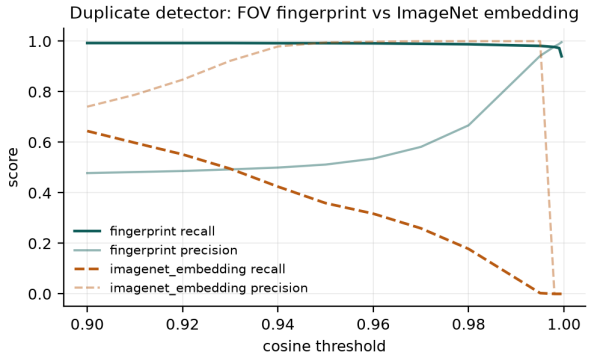


Fig. 4. Duplicate-detector calibration: rejected embedding method vs. FOV fingerprint.

substantial downscale that ImageNet features are not invariant to. An FOV-normalized 64×64 pixel fingerprint recovers **97%** of the same ground truth at 0.99 precision and is used throughout the rest of this paper (Fig. 4).

The detector's threshold required a correction after an initial miscalibration: calibrating on a synthetic balanced sample suggested 0.98 was safe, but applied to the true search space (positive rate on the order of one in a million pairs) it returned 26,634 false-positive pairs, chaining into duplicate clusters covering 37% of the dataset. Recalibrating against hard negatives drawn from the real search space gave the 0.999 operating point reported above. Within ODIR, this detector found 2 duplicate groups (5,058 → 5,056 images); across the two benchmarks, 1,626 merged-benchmark images were identified as ODIR copies and excluded from the cross-source transfer evaluation (Section IV-D).

IV. RESULTS

A. Protocol Ladder: Patient Leakage on a Frozen Classifier

Holding a frozen-ResNet50-features-plus-logistic-head model fixed, four splitting protocols were compared.

No rung differs significantly from any other (McNemar $p \in [0.14, 0.75]$; all confidence intervals overlap; Fig. 5). Read alone, this falsifies the hypothesis that patient leakage inflates accuracy on this benchmark. It is reported as a lower bound rather than a conclusion, because a frozen linear head has essentially no capacity to memorize a specific training example — and leakage inflation specifically requires memorization.

TABLE IV
PROTOCOL LADDER, FROZEN FEATURES

Rung	n	Macro-F1	95% CI
Image-level	5,058	0.5496	[0.5308, 0.5677]
Patient-disjoint	5,058	0.5444	[0.5248, 0.5622]
+ Dedup	5,056	0.5476	[0.5282, 0.5675]
Provenance-controlled	1,535	0.5048	[0.4735, 0.5383]

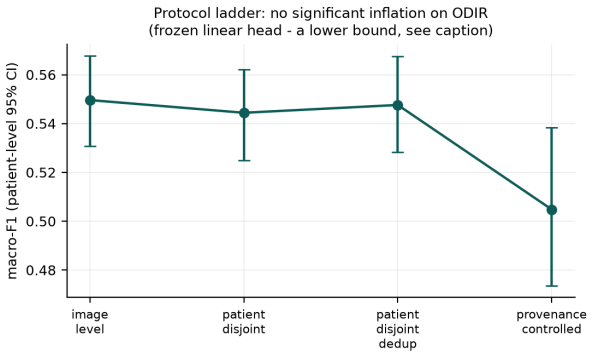


Fig. 5. Protocol ladder across four splitting rungs.

TABLE V
LEAKAGE INFLATION VS. MODEL CAPACITY

Model	Capacity	Inflation	McNemar p
Logistic	none	+0.0052	0.7467
MLP	low	+0.0274	0.2006
5-NN	moderate	+0.0207	0.00065
1-NN	pure memorizer	+0.0453	∼0

TABLE VI
FINE-TUNED CONFIRMATION (E3)

Rung	n	Macro-F1	95% CI
Image-level (leaky)	5,058	0.6797	[0.6611, 0.6967]
Patient-disjoint + dedup	5,056	0.6579	[0.6393, 0.6768]

B. Capacity Ladder: Does Inflation Require Memorization?

Holding features and splits fixed, model capacity was swept from a logistic head up to a pure 1-nearest-neighbor (1-NN) retriever, which recalls a leaked fellow-eye example by construction whenever one is present in the training set.

Inflation increases monotonically with capacity and becomes statistically significant once the model can meaningfully memorize (5-NN, 1-NN). This corrects the interpretation of the protocol ladder: the earlier null result was an artifact of testing an architecture that structurally cannot exploit patient leakage, not evidence the benchmark is free of it (Fig. 6).

C. Confirmation on a Fine-Tuned, Deployable Model

To confirm the mechanism on a model that would actually be deployed, a ResNet50 was fine-tuned end-to-end (8 epochs, seed 42, 384×384 input) on a GPU, on the image-level and patient-disjoint-deduplicated rungs.

TABLE VII
INFLATION BY CAPACITY, INCLUDING FINE-TUNED CONFIRMATION

Model	Capacity	Inflation	p
Logistic	none	+0.0052	0.75
Fine-tuned ResNet50	**realistic**	**+0.0218**	**0.0007**
1-NN retriever	pure	+0.0453	∼0

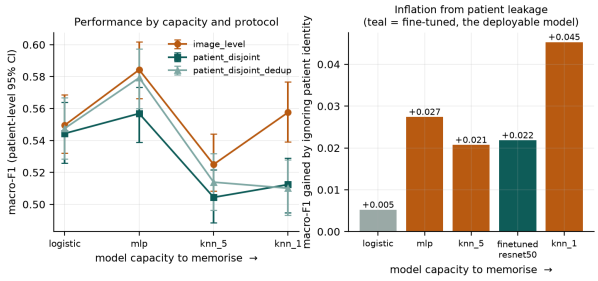


Fig. 6. Capacity ladder, including the fine-tuned ResNet50 confirmation.

TABLE VIII
CROSS-SOURCE TRANSFER

Source	n	Classes	Majority acc.	Model acc.
`crop_jpg256`	1,223	C, G	0.517	0.330
`eyepacs_jpeg`	1,098	D	1.000	0.297
`odir_jpg512`	69	C, G, N	0.449	0.507
`png_highres`	201	C, G	0.502	0.104

Inflation: $+\mathbf{0.0218}$ macro-F1 (McNemar $p = 0.0006825$, 886 discordant predictions across 5,056 paired test cases).

The pattern remains monotonic in capacity and is now significant on the model a screening application would actually use. This upgrades the claim from a proxy-model demonstration to a confirmed property of a deployable classifier: patient-disjoint splitting matters, and matters more as model capacity increases — precisely the direction the field's architectures are moving. The result should be read as confirmatory rather than exhaustively validated: it reflects one random seed and one backbone trained for eight epochs, and the seed-stability treatment (Section IV-F) was not repeated for this fine-tuned run.

D. Cross-Source Transfer

The frozen-feature model trained on the 5,056 deduplicated ODIR images was evaluated on merged-benchmark images from four acquisition sources, after excluding the 1,626 confirmed ODIR copies, so no test image had been seen during training.

On three of four unseen sources, model accuracy falls below simply guessing the majority class for that source (Fig. 7). Because each source is class-restricted — itself a symptom of the provenance confound — results are reported per source rather than pooled. Two caveats apply: the in-domain model is itself relatively weak (macro-F1 0.55 on frozen features), so part of this failure reflects model quality rather than a domain gap specifically; and this measures transfer under

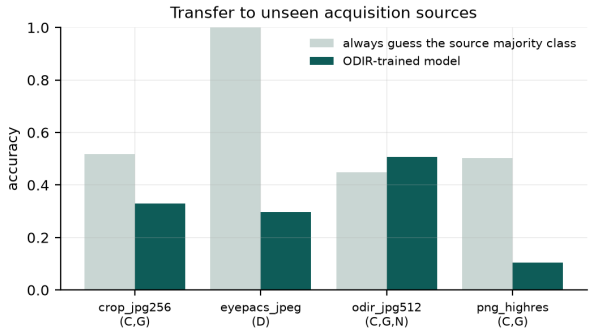


Fig. 7. Cross-source transfer to four unseen acquisition sources.

TABLE IX
IMAGE-MODEL PERFORMANCE UNDER DEMOGRAPHIC CONTROLS

Condition	n	Macro-F1	95% CI	Cataract AUROC
Unrestricted	5,056	0.5476	[0.528, 0.567]	0.9462
Age-matched	964	0.5385	[0.503, 0.572]	0.9414
Sex-matched	972	0.5302	[0.500, 0.562]	0.9406
Multi-finding removed	4,679	0.5478	[0.526, 0.568]	0.9560

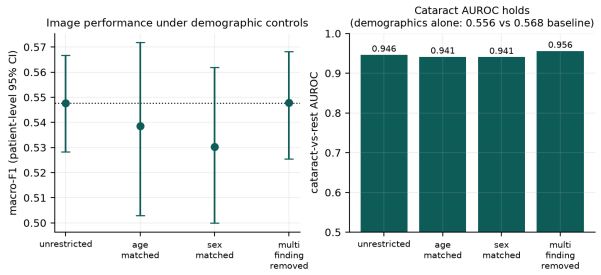


Fig. 8. Model performance under age/sex matching.

a documented, discrete change of acquisition source, not generalization in an unqualified sense.

E. Disease Detector or Demographics Detector?

Because convolutional networks can read age and sex from fundus photographs, the demographic confound documented in Section III was tested directly rather than only disclosed. A classifier given age and sex alone, with no image, scored 0.556 accuracy against a 0.568 majority-class baseline (macro-F1 0.2531) — **worse** than always guessing the majority class.

Matching was performed by downsampling within strata, closing the demographic route without reweighting or modelling assumptions (Fig. 8). Every confidence interval under a control overlaps the unrestricted interval, and cataract-versus-rest AUROC remains near 0.94 throughout. **The demographic confound is real in the data, but it is not what the image model is using** — the strongest single piece of evidence in this paper that classifier accuracy reflects retinal content rather than a shortcut.

TABLE X
SEED STABILITY

Model	Mean inflation	sd	Range across seeds
1-NN	+0.0453	0.0027	[+0.0434, +0.0484]
Logistic	−0.0031	0.0085	[−0.0126, +0.0038]

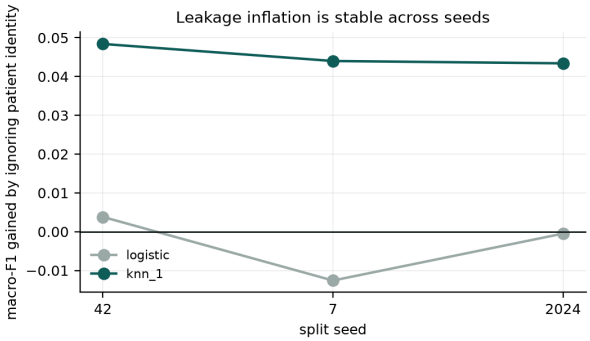


Fig. 9. Leakage inflation across three random split seeds.

F. Seed Stability and Multiple Comparisons

Because a finding that holds under only one random split is not a finding, the leakage comparison was re-run under three independent split seeds (42, 7, 2024).

The memorizer's effect is large, tightly reproducible, and significant at every seed (Fig. 9). The logistic head's effect is small, noisy, and **changes sign between seeds** — a direct illustration that this effect is statistical noise rather than a real property of the leakage mechanism at low model capacity. Because this study performs 13 paired hypothesis tests, a Holm–Bonferroni correction was applied across the full family at $\alpha = 0.05$. Five of thirteen tests survive correction, and all five are memorization-based results. One test in particular, *leakage[logistic, seed 7]*, had an uncorrected $p = 0.037$ and a *negative* effect direction — it would have been reported as significant without correction, and would have been wrong. This case is the concrete justification for applying the correction.

V. DOCUMENT PROCESSING

A. Corpus and Extraction

Because no suitable public de-identified ophthalmic referral corpus was located — a search that is recorded here rather than silently substituted — a synthetic corpus of 360 referral documents was generated across six visual layouts (A–F). Two extraction methods were developed and tuned using only layouts A–C; layouts D–F were sealed and opened exactly once, after both extractors were frozen, following the held-out-layout convention established by DocILE [10].

Both extractors degrade on sealed layouts relative to development layouts (Fig. 10), which is expected of any layout-sensitive extraction pipeline and motivates estimating extraction reliability rather than assuming it (Section V-B). Every number in this section carries its synthetic-data caveat: this

TABLE XI
EXTRACTION PERFORMANCE

Extractor	Split	Prec.	Rec.	F1	Exact
E1	dev	0.696	0.534	0.604	0.568
E1	sealed	0.726	0.235	0.355	0.297
E2	dev	0.687	0.679	0.683	0.691
E2	sealed	0.563	0.430	0.488	0.453

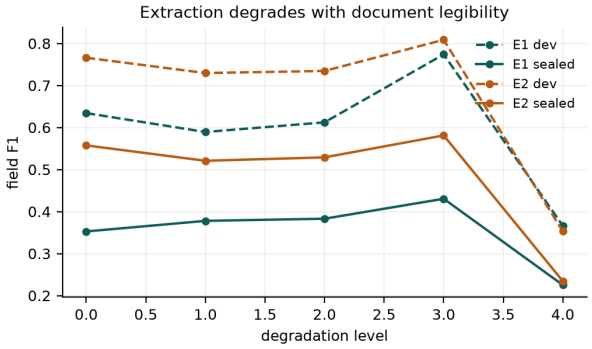


Fig. 10. Extraction performance by layout degradation.

TABLE XII
RELIABILITY ESTIMATOR COMPARISON

Extractor	Signal	AUROC	Brier	ECE
E1	raw OCR conf.	0.5214	0.2473	0.2254
E1	learned estimator	0.9799	0.0542	0.0294
E2	raw OCR conf.	0.6389	0.3806	0.3854
E2	learned estimator	0.9871	0.0427	0.0151

is a technical evaluation under controlled degradation, not a clinical validation.

B. Reliability of the Extraction Signal

Raw OCR engine confidence and a small learned estimator (a gradient-boosted classifier trained on extraction-level features) were compared as predictors of whether an extracted clinical field was correct, evaluated on sealed layouts.

Raw OCR confidence is close to chance-level for E1 and only modestly better for E2 (Fig. 11) — it measures how clearly the engine read the text, not whether the resulting value is correct for its field. The learned estimator's feature importances explain the gap: value format-validity and value length dominate the decision (combined weight ≈ 0.86), while raw OCR confidence contributes ≈ 0.08. A field can be read with high OCR confidence and still be semantically wrong for its slot, which is exactly the failure raw confidence cannot detect and a format check can.

VI. TRIAGE

A. Deferral with Document Evidence

The screening prototype answers with a predicted class whenever it chooses to answer, and defers otherwise. Three systems were compared against an image-confidence-only baseline: System A, two hand-written rules combining image

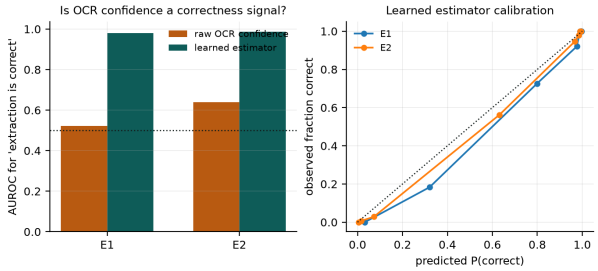


Fig. 11. Raw OCR confidence vs. learned extraction-correctness estimator.

TABLE XIII
DEFERRAL SYSTEM COMPARISON

System	AURC ↓	Error-det. AUROC	Acc.@50%
Image conf. only	0.3889	0.5875	0.5889
System A (rules)	**0.1973**	**0.8817**	0.8667
System B (learned)	0.2107	0.8692	0.8889
System B w/o $\hat{c}$	0.2032	0.8705	0.8667

confidence and document agreement; System B, a learned gate that additionally takes the extraction-reliability estimate $\hat{c}$ (Section V-B) as input; and System B without $\hat{c}$, isolating that estimate's contribution. A falsification rule was fixed before this experiment ran: if System B did not outperform System A, the paper would present the rule-based system and report the learned gate as a negative result.

Document evidence produces a substantial improvement over image confidence alone (Fig. 12) — AURC falls from 0.389 to approximately 0.20, and error-detection AUROC rises from 0.59 to 0.88. That improvement, however, is delivered by the **hand-written rules**, not the learned gate: System B does not outperform System A, and removing $\hat{c}$ from System B *improves* it slightly rather than degrading it. Per the pre-registered rule, this paper presents System A as the deferral mechanism and reports the learned gate as a negative result rather than a headline contribution. $\hat{c}$ is not thereby shown to be useless — it remains a strong estimator of extraction *correctness* on its own terms (Section V-B, AUROC 0.98–0.99) — but it adds nothing measurable to the deferral decision once image confidence and document agreement are available. This comparison rests on 180 sealed-layout rows; a $\approx$0.01 AURC gap between System B variants is within the noise that sample size implies, so the accurate statement is that the learned gate did not demonstrate an improvement, not that the rules are proven superior.

B. Robustness Across Evidence Conditions

A robustness matrix (Fig. 13) varies referral-document quality and availability and confirms the deferral advantage from document evidence persists across degraded-document conditions.

C. Attribution

Grad-CAM++, implemented via forward and backward hooks without an external library dependency, was computed

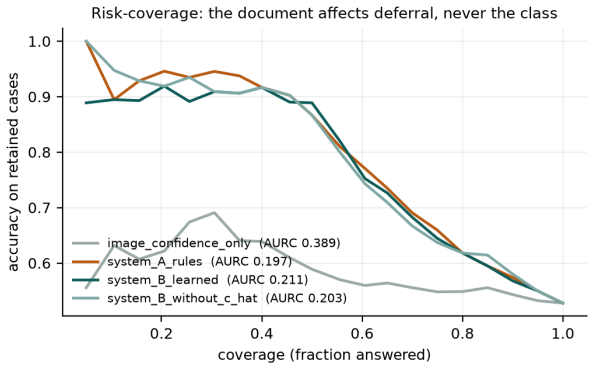


Fig. 12. Risk-coverage curves for all four deferral systems.

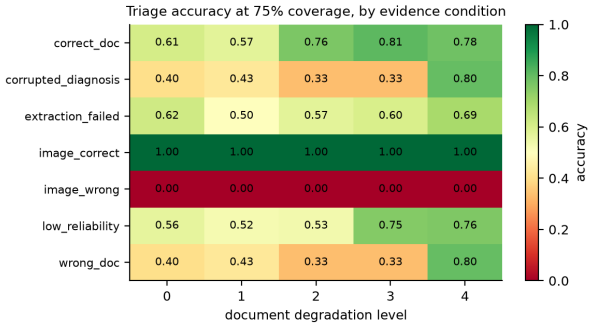


Fig. 13. Deferral robustness across document-quality conditions.

TABLE XIV
ATTRIBUTION SUMMARY

Measure	Value
FOV-mass ratio (observed)	0.8364
Uniform-attribution baseline	0.5348
Concentration ratio	1.56×
Deletion AUC (mean)	0.4687
Randomization sanity check	pass ($r = -0.0943$)

over the frozen-feature model on a held-out fold the classification head had not been fitted on.

Attribution concentrates within the retinal field of view at 1.56× what uniform attention would produce (Fig. 14), and the randomization sanity check confirms the maps depend on the model's learned weights rather than the attribution method's own structure. This licenses the statement that attribution falls on retinal tissue rather than image borders or letterboxing; it does not license any claim that the model attends to a specific lesion, since no lesion masks and no ophthalmologist review were available. Attribution here was computed on the frozen-feature model rather than the fine-tuned model from Section IV, since Grad-CAM requires gradients through the backbone being evaluated, and re-running this analysis on the fine-tuned backbone was not performed in this iteration of the project.

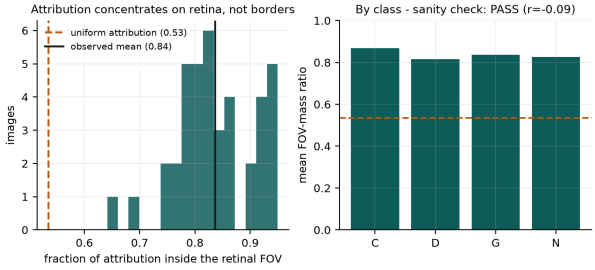


Fig. 14. Grad-CAM++ attribution and randomization sanity check.

VII. LIMITATIONS

- Cataract and glaucoma classes contain 203 and 182 patients respectively. Findings involving these classes are reported as confidence intervals, not point estimates, and out-of-fold predictions are pooled before any significance test.
- ODIR is less confounded than a naive multi-source merge, not confound-free: an ≈11-year age gap separates cataract and DR patients, sex distribution differs by class, and 23.2% of glaucoma-labelled rows name a second condition.
- The protocol-ladder result (Section IV) used a frozen linear head, which structurally cannot memorize. Section IV-B addresses this by sweeping model capacity, and the fine-tuned confirmation addresses it directly — but every model in those sections still sits on frozen ImageNet features, and a differently fine-tuned backbone could behave differently again.
- Attribution (Section VI) is computed for the frozen-feature model, not the fine-tuned diagnostic model. It demonstrates attribution is concentrated on retinal tissue and passes a randomization sanity check; it does not demonstrate the model is examining a lesion.
- The referral corpus is synthetic. Every derived number is a technical evaluation under controlled degradation, never a clinical validation. A search for a public, de-identified ophthalmic referral corpus is recorded rather than substituted with an unstated proxy.
- The duplicate detector's recall is measured over ODIR↔ merged-benchmark ground-truth pairs specifically; recall for duplicates occurring entirely within ODIR is not directly measurable with the ground truth available.
- This work involves no prospective data collection, no clinical trial, and no regulatory clearance of any kind.

VIII. CONCLUSION

This paper set out to measure, rather than assume, how much of a widely reported four-class ophthalmic screening benchmark's performance reflects retinal pathology as opposed to how the benchmark was assembled. Acquisition provenance was found to explain a substantial share of separability, almost entirely for the diabetic-retinopathy and normal classes and considerably less for cataract versus glaucoma. Patient-level

leakage produced no measurable effect on a low-capacity classifier but a significant, monotonic effect that grows with model capacity, confirmed directly on a fine-tuned deployable model rather than inferred from a proxy. The trained classifier's apparent competence was shown not to rest on the dataset's demographic confounds, and was shown not to transfer to unseen acquisition sources. On the document-processing side, raw OCR confidence was shown to be an unreliable signal of extraction correctness, a learned alternative closed almost all of that gap, and a pre-registered falsification rule led this paper to present hand-written deferral rules rather than a more elaborate learned gate that did not, in fact, perform better. Every finding above is reported with the model, sample size, and statistical test that produced it, and every negative result — the initial leakage null, the failed embedding-based duplicate detector, the learned deferral gate — is reported as a negative result rather than omitted.

REFERENCES

[1] A. Torralba and A. A. Efros, "Unbiased look at dataset bias," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2011, pp. 1521–1528.

[2] Z. Liu and K. He, "A decade's battle on dataset bias: Are we there yet?" *arXiv preprint arXiv:2403.08632*, 2024.

[3] I. E. Tampu, A. Eklund, and N. Haj-Hosseini, "Inflation of test accuracy due to data leakage in deep learning-based classification of OCT images," *Scientific Data*, vol. 9, 2022, doi: 10.1038/s41597-022-01618-6.

[4] C. Leibig, V. Allken, M. S. Ayhan, P. Berens, and S. Wahl, "Leveraging uncertainty information from deep neural networks for disease detection," *Scientific Reports*, vol. 7, no. 17816, 2017.

[5] T. Araújo, G. Aresta, L. Mendonça, S. Penas, C. Maia, Â. Carneiro, A. M. Mendonça, and A. Campilho, "DR|GRADUATE: Uncertainty-aware deep learning-based diabetic retinopathy grading in eye fundus images," *Medical Image Analysis*, vol. 63, p. 101715, 2020.

[6] Q. Zhang, H. Wu, C. Zhang, Q. Hu, H. Fu, J. T. Zhou, and X. Peng, "Provable dynamic fusion for low-quality multimodal data," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023.

[7] Z. Han, C. Zhang, H. Fu, and J. T. Zhou, "Trusted multi-view classification," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[8] J. She and K. Spivakovsky, "MultiRetNet: A multimodal vision model and deferral system for staging diabetic retinopathy," *arXiv preprint arXiv:2507.14738*, 2025.

[9] A. Yavari, T. Schmoll, R. A. Leitgeb, K. L. Huber, H. Stino, A. Pollreisz, W. Drexler, and T. Schlegl, "Applicability study of AI attribution methods for ophthalmic image classification," *Scientific Reports*, 2026, doi: 10.1038/s41598-025-33120-5.

[10] Š. Šimsa, M. Šulc, M. Uřičář, Y. Patel, A. Hamdi, M. Kocián, M. Skalický, J. Matas, A. Doucet, M. Coustaty, and D. Karatzas, "DocILE benchmark for document information localization and extraction," in *Proc. Int. Conf. Document Analysis and Recognition (ICDAR)*, 2023.